

Othmane Mazhar

POSTDOCTORAL RESEARCHER · PROBABILITY, MATHEMATICAL STATISTICS, AND OPTIMAL TRANSPORT

Institut Louis Bachelier — Fondation du Risque, Chair in Artificial Intelligence and Quantitative Methods for Finance, Paris, France

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Research Profile

I develop mathematical and statistical foundations for learning, generating, and controlling stochastic systems from dependent data. My work combines finite-sample statistical theory, optimal transport, stochastic control, and computational PDE methods. My programme focuses on Schrödinger–Bass transport, learning stochastic bridge dynamics, path-space probability, and quantitative stability guarantees for generated path laws.

Research areas Mathematical statistics; stochastic control; optimal transport; financial engineering; stochastic generative models; scientific machine learning

Academic Appointments

Institut Louis Bachelier

Paris, France

Postdoctoral Researcher

2026–Present

- Hosted within Fondation du Risque. Scientific supervisor: Prof. Nizar Touzi, Professor and Chair of Finance and Risk Engineering at NYU Tandon School of Engineering. Industry-funded research position supported by Jump Trading.
- Developing the mathematical foundations of Schrödinger–Bass transport for synthetic financial time-series generation, including multi-marginal extensions, convergence and stability of training schemes, and statistical error propagation.

Laboratoire de Probabilités, Statistique et Modélisation

Paris, France

Postdoctoral Researcher

2023–2026

- Joint research laboratory of Sorbonne Université and Université Paris Cité. Scientific supervisor: Prof. Huyên Pham, École polytechnique. Industry-funded research programme supported by BNP Paribas.
- Initiated and led the programme on direct statistical estimation of Schrödinger-bridge dynamics: formulated the estimation problem, derived the conditional-ratio representation underlying the method, constructed the estimators, and developed their finite-sample, adaptive, asymptotic, and minimax theory.
- Developed the noncoercive duality and attainment analysis for semimartingale Schrödinger–Bass transport, the reduced-dual formulation over terminal potentials, and the verification arguments connecting dual optimizers with the optimal controlled diffusion.

KTH Royal Institute of Technology

Stockholm, Sweden

Graduate Teaching Assistant

2017–2023

- Appointments in the Department of Mathematics and the School of Electrical Engineering and Computer Science.

Education

KTH Royal Institute of Technology

Stockholm, Sweden

PhD in Applied and Computational Mathematics: Probability and Statistics

2017–2022

- Thesis: *Data-Driven Modeling in the Presence of Time Series Structure: Improved Bounds and Effective Algorithms*.
- Advisor: Boualem Djehiche.

KTH Royal Institute of Technology

Stockholm, Sweden

MSc in Applied and Computational Mathematics

2014–2016

- Perfect GPA (4.0/4.0); KTH Scholarship for Academic Excellence.
- Thesis: *Optimal Consensus and Opinion Dynamics*. Advisor: Xiaoming Hu.

- Highest GPA and Best Master's Thesis Award.
- Thesis: *Statistical Control for a Silver Production Process*. Advisor: Loubna Benabbou.

Publications and Manuscripts

Peer-Reviewed Journal Articles

- S. E. Choutri, B. Djehiche, **O. Mazhar**. "Empirical validation of novel non-asymptotic bounds on the least squares estimator for LTI systems with applications in economics." *Asian Journal of Economics and Banking* 9(3) (2025), 491–509. doi:10.1108/AJEB-12-2024-0147.
- S. Shi, **O. Mazhar**, B. De Schutter. "Finite-sample analysis of identification of switched linear systems with arbitrary or restricted switching." *IEEE Control Systems Letters* 7 (2023), 121–126. doi:10.1109/LCSYS.2022.3187511.
- B. Djehiche, **O. Mazhar**, C. R. Rojas. "Finite impulse response models: A non-asymptotic analysis of the least squares estimator." *Bernoulli* 27(2) (2021), 976–1000. doi:10.3150/20-BEJ1262.

Peer-Reviewed Conference Papers

- S. E. Choutri, P. Chauhan, **O. Mazhar**, S. E. Jabari. "Monte Carlo-Type Neural Operator for Differential Equations." *IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2025. arXiv:2510.05620.
- A. Athrey, **O. Mazhar**, M. Guo, B. De Schutter, S. Shi. "Regret analysis of learning-based linear quadratic Gaussian control with additive exploration." *European Control Conference (ECC)*, 2024, 1795–1801.
- O. Mazhar**, C. R. Rojas, C. Fischione, M. R. Hesamzadeh. "Bayesian Model Selection for Change Point Detection and Clustering." *Proceedings of the 35th International Conference on Machine Learning*, PMLR 80 (2018), 3433–3442.

Book Chapter

- B. Djehiche, **O. Mazhar**. "An optimal non-asymptotic journey through system identification with least squares." In *Artificial Intelligence and Machine Learning for Econometrics: Applications, Regulation (and Related Topics)*, Studies in Systems, Decision and Control. Springer, 2025.

Preprints and Manuscripts

- O. Mazhar**. "A Constructive Solution to Talagrand's Gaussian Convexification Problem." Preprint, 2026. arXiv:2609.14832.
- O. Mazhar**, H. Pham. "Direct Estimation of Schrödinger Bridge Time-Series Drifts: Finite-Sample, Asymptotic, and Adaptive Guarantees." Under review, 2026. arXiv:2605.05432.
- P. Henry-Labordère, G. Loeper, **O. Mazhar**, H. Pham, N. Touzi. "Bridging Schrödinger and Bass: A Semimartingale Optimal Transport Problem with Diffusion Control." Under review, 2026. arXiv:2603.27712.
- A. Alouadi, G. Loeper, C. Marsala, **O. Mazhar**, H. Pham. "SBBTS: A Unified Schrödinger–Bass Framework for Synthetic Financial Time Series." Under review, 2026. arXiv:2604.07159.
- A. Alouadi, P. Henry-Labordère, G. Loeper, **O. Mazhar**, H. Pham, N. Touzi. "LightSBB-M: Bridging Schrödinger and Bass for Generative Diffusion Modeling." Under review, 2026. arXiv:2601.19312.
- A. Alouadi, P. Henry-Labordère, G. Loeper, **O. Mazhar**, H. Pham, N. Touzi. "A PDE Derivation of the Schrödinger–Bass Bridge." Under review, 2026. arXiv:2601.17863.
- B. Djehiche, **O. Mazhar**. "Minimax Lower Bounds and Adaptivity for LTI State-Space Models with van Trees." Under review, 2025.
- B. Djehiche, **O. Mazhar**. "Efficient Learning of Hidden-State LTI State-Space Models of Unknown Order." Under review, 2025. arXiv:2202.01625.

Theses and Earlier Preprint

- O. Mazhar**. *Data-Driven Modeling in the Presence of Time Series Structure: Improved Bounds and Effective Algorithms*. PhD thesis, KTH Royal Institute of Technology, 2022.
- O. Mazhar**. *Optimal Consensus and Opinion Dynamics*. Master's thesis, KTH Royal Institute of Technology, 2016. Advisor: Xiaoming Hu.
- B. Djehiche, **O. Mazhar**. "Non-asymptotic estimation lower bounds for LTI state-space models with Cramér–Rao and van Trees." Earlier preprint, 2021. arXiv:2109.08582. Substantially extended in the subsequent manuscript on minimax lower bounds and adaptivity.

Selected Research in Progress

Gaussian Schrödinger–Bass Bridges

Joint work with Huyền Pham. Developed the mathematical analysis of the unrestricted Gaussian stochastic-control problem: proved that the optimal reduced-dual potential is necessarily quadratic, established the unrestricted-dual justification connecting the finite-dimensional Gaussian system with the full stochastic-control problem, and derived the exact mean–covariance decomposition. Current work develops the resulting geodesic structure and limiting regimes connecting Schrödinger and Bass transport.

A Continuous Version of Talagrand’s Gaussian Convexification Problem

Joint work with Nizar Touzi extending the constructive Gaussian convexification framework to a continuous formulation.

Learning Schrödinger-Bridge Dynamics from Dependent Data

Extension of the direct drift-estimation programme from independent endpoint observations to a single dependent trajectory, with finite-sample theory under quantitative assumptions on mixing, recurrence, local mass, smoothness, and tails.

Nonlinear Path-Space Transport–Information Inequalities

Solo project developing transport–information inequalities and quantitative stability estimates for stochastic systems when both drift and volatility vary, with applications to mean-field diffusions, interacting particle systems, and empirical path measures.

Learning Corrections for PDE Solvers

Developing learning-based error and correction spaces for numerical PDE approximations, together with residual-based correction and instance-wise certification.

Selected Talks and Presentations

2025	An Optimal Interpolation Diffusion Model for Time Series: Bridging Schrödinger and Bass , Invited speaker, LPSM Workshop on Financial and Actuarial Mathematics and Numerical Probability	Paris, France
2024	Bridging Schrödinger and Bass , Invited speaker, Workshop on Stochastic Control, Optimal Transport, Calibration, and Applications in Finance	Carry-le-Rouet, France
2019	Finite Impulse Response Models in Low and High Dimension , Poster, ERNSI Workshop	Maastricht, Netherlands
2018	Bayesian Model Selection for Change Point Detection and Clustering , Oral presentation, International Conference on Machine Learning	Stockholm, Sweden
2018	Multiple Regime-Switching Regression for Electricity Spot-Price Prediction , Poster, ERNSI Workshop	Cambridge, UK
2017	Model Selection for Change Point Detection and Clustering , Poster, ERNSI Workshop	Sainte-Foy-lès-Lyon, France

Teaching Experience

High-Dimensional Probability and Statistics for Data Science

KTH

Organizer and Lecturer

2021

- At the request of colleagues, independently developed and delivered a non-curricular internal series of fourteen two-hour lectures on non-asymptotic probability and high-dimensional statistics.
- Topics included concentration inequalities, random vectors, random matrices, chaining, and applications to statistical learning.

Statistical Machine Learning

KTH

Organizer and Lecturer

2020

- At the request of research colleagues, independently designed and delivered a series of eight one-hour lectures on statistical learning theory and modern machine-learning methods.
- Topics included statistical learning, kernel methods, and large-scale and stochastic optimization.

EL2820 Modelling of Dynamical Systems

KTH

Teaching Assistant

2018–2020

- Delivered nine two-hour exercise sessions per offering and supervised laboratory and computer-lab components for large en-

- gineering cohorts, including computer-lab sessions with approximately 100 students.
- Redesigned computer-lab material and contributed to exam preparation and grading across multiple course offerings.

Student Supervision and Research Mentoring

Bachelor's Thesis Supervision

2020	Daniel Hirsch and Tim Steinholtz, Bachelor's thesis supervision	KTH
2019	Emil Backman and David Petersson, Bachelor's thesis supervision	KTH
2019	David Mainwaring and Jonathan Österberg, Bachelor's thesis supervision	KTH
2018	Mikaela Åstrand and Oscar Bergqvist, Bachelor's thesis supervision	KTH

Research Mentoring

Alexandre Alouadi, PhD Candidate

École polytechnique

Research Mentor and Collaborator

2025–2026

- Collaborated regularly on two projects forming part of his doctoral research.
- Contributed to problem formulation, theoretical development, methodology, experimental design, and preparation of joint manuscripts.

Archith Athrey

TU Delft

Master's Thesis Co-Supervisor, Research Mentor, and Collaborator

2023–2024

- Co-supervised his master's research, contributing to problem formulation, theoretical development, methodology, and analysis.
- The collaboration continued beyond the thesis and led to a joint ECC publication on learning-based linear-quadratic Gaussian control.

Carl-Johan Larsson and Fredrik Håkansson

KTH / Lund
University / Spotify

Industrial Master's Thesis Supervisor and Research Mentor

2018–2019

- Supervised the joint thesis *User-Based Predictive Caching of Streaming Media*, undertaken by two experienced software engineers completing master's studies alongside their professional careers.
- Primarily mentored the machine-learning component, including problem formulation, recurrent-neural-network modelling, trajectory prediction, and evaluation using large-scale mobility data.
- The project integrated predicted vehicle trajectories with cellular-coverage information to trigger predictive media caching before connectivity loss.

Awards, Fellowships, and Grants

2015	KTH Scholarship for Academic Excellence, KTH Royal Institute of Technology	SEK 180,000
2013	Best Master's Thesis Award, École Mohammadia d'Ingénieurs	
2013	Highest GPA Award, École Mohammadia d'Ingénieurs	

Academic Service and Professional Development

Peer review	Artificial Intelligence and Statistics (AISTATS); Conference on Learning Theory (COLT); International Conference on Machine Learning (ICML); Conference on Neural Information Processing Systems (NeurIPS)
Reviewing recognition	Gold Technical Reviewer, International Conference on Machine Learning (ICML), 2026
Conference service	Volunteer organizer, Conference on Learning Theory (COLT), Stockholm, 2018

Selected Schools and Workshops

2025	Optimal Transport: From Robust Pricing to Model Calibration , Dolomites Winter School	<i>Folgarida, Italy</i>
2025	Advances in Stochastic Control and Reinforcement Learning , BIRS workshop organized by NYU Paris	<i>Paris, France</i>
2024	Stochastic Control, Optimal Transport, Calibration, and Applications in Finance , Workshop and winter school	<i>Carry-le-Rouet, France</i>
2021	The Interplay between High-Dimensional Geometry and Probability , Hausdorff Research Institute for Mathematics	<i>Bonn, Germany</i>
2019	High-Dimensional Probability and Algorithms , PSL Mathematics Summer School	<i>Paris, France</i>

Technical Skills and Languages

Programming	Python; C/C++; Go; MATLAB
Languages	English; Arabic; French
Citizenship	Swedish; Moroccan

References

Available upon request.